

Linear Models: Regression

Machine Learning (Bilingual)

齐飞 (Fei **Qi**) <fred.qi@ieee.org>

李阳 (Yang **Li**)

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School of Artificial Intelligence

Xidian University





Univariate Linear Regression

Multivariate Linear Regression

Features and Generalized Linear Regression

Regularization

Bayesian Linear Regression

Univariate Linear Regression



Supervised Learning

Given the “right answer” for each example in the data.



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Regression

To predict real-valued output.

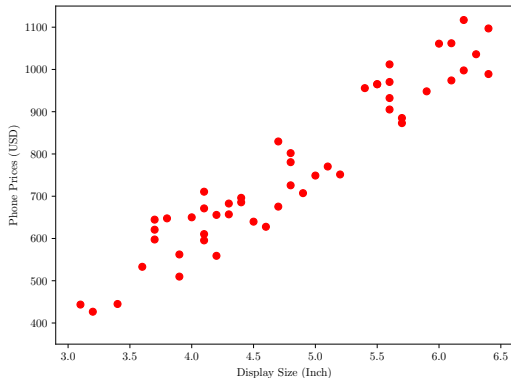
Supervised Learning

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To predict real-valued output.

Smart Phone Price Prediction



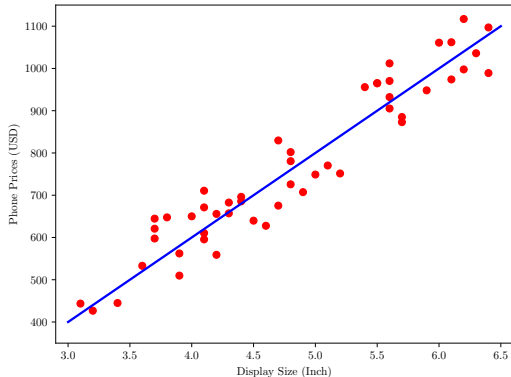
Supervised Learning

Given the “right answer” for each example in the data.

Regression

To predict real-valued output.

Smart Phone Price Prediction





Training set of phone prices

Display size (x)	Price (USD) (y)
4.6	627.7
4.7	829.6
5	748.9
6.4	989.1
4.4	685.6
$\vdots$	$\vdots$



Training set of phone prices

Display size (x)	Price (USD) (y)
4.6	627.7
4.7	829.6
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4.4	685.6
$\vdots$	$\vdots$

Notations

- m Number of training examples
- x input variable/features
- y output variable



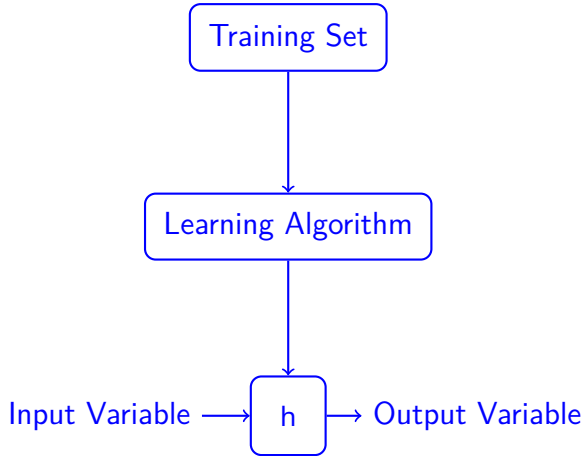
Training set of phone prices

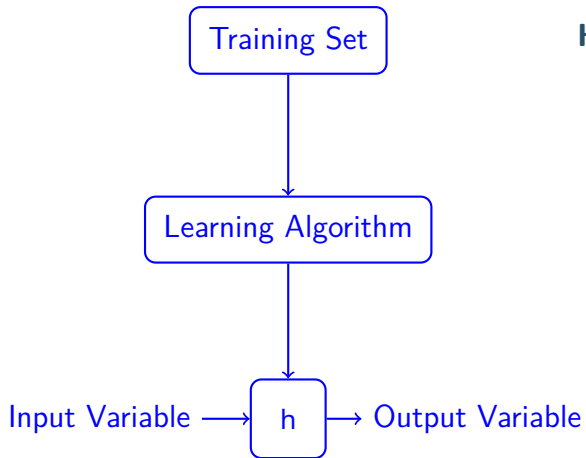
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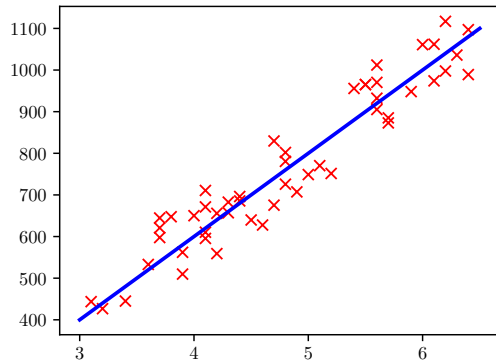
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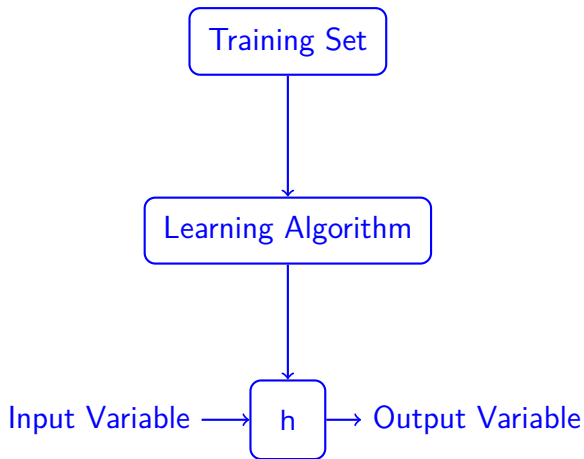
- $x^{(1)} = 4.6$, $y^{(1)} = 627.7$
- $x^{(2)} = 4.7$, $y^{(2)} = 829.6$
- ...
- $x^{(m)}$, $y^{(m)}$



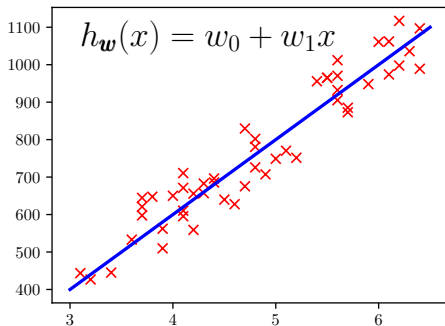


How do we represent h ?





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- Regression with one variable
- **Univariate** regression



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⋮	⋮

- Hypothesis: $h_w = w_0 + w_1x$
- w_i 's: parameters
- How to choose w_i 's?



- Hypothesis

$$h_{\mathbf{w}}(x) = w_0 + w_1x$$

- Parameters w_0, w_1

- Cost Function

$$J(w_0, w_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right)^2$$

- Goal

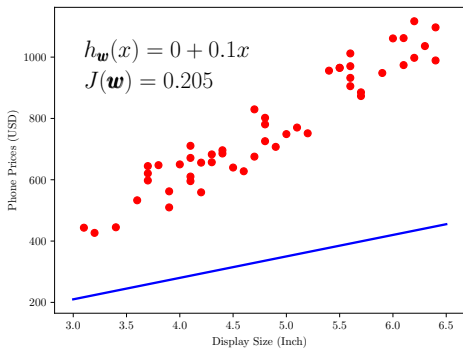
$$\arg \min_{w_0, w_1} J(w_0, w_1)$$

An example



$$h_{\mathbf{w}}(x)$$

For fixed w_0, w_1 , this is a function of x .

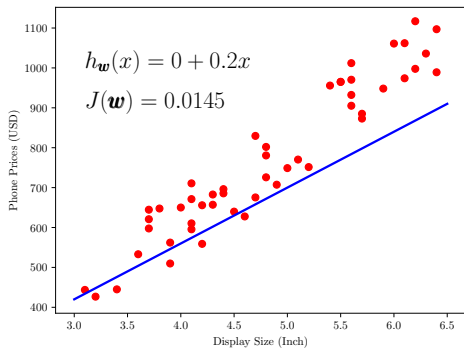


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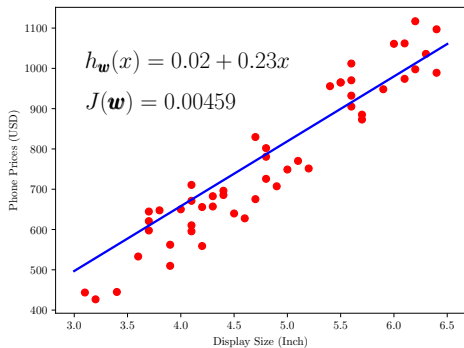


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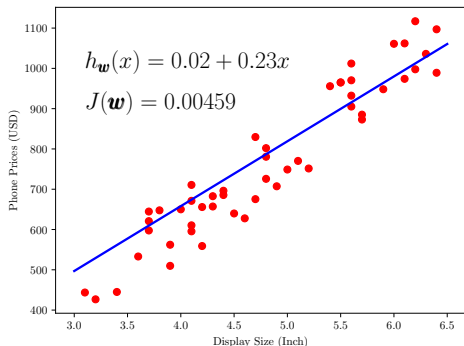


An example



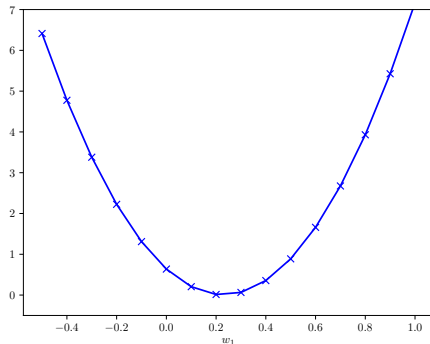
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$$J(\mathbf{w}_0, \mathbf{w}_1)$$

Functions of the parameters w_0, w_1 .





- Have some function $J(w_0, w_1)$
- Want $\arg \min_{w_0, w_1} J(w_0, w_1)$



- Have some function $J(w_0, w_1)$
- Want $\arg \min_{w_0, w_1} J(w_0, w_1)$

Outline

- Start with some w_0, w_1
- Keep changing w_0, w_1 to reduce $J(w_0, w_1)$
- until we end up at a minimum

Gradient descent algorithm



repeat until convergence {

$$w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(w_0, w_1) \quad (\text{for } j = 0 \text{ and } j = 1)$$

}

α : Learning rate

Gradient descent algorithm



repeat until convergence {
 $w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(w_0, w_1)$ (for $j = 0$ and $j = 1$)
}

α : Learning rate

- Simultaneous update

$$\text{temp0} := w_0 - \alpha \frac{\partial}{\partial w_0} J(w_0, w_1)$$

$$\text{temp1} := w_1 - \alpha \frac{\partial}{\partial w_1} J(w_0, w_1)$$

$$w_0 := \text{temp0}$$

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- Incorrect

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Comments on gradient descent



Gradient descent can converge to a local minimum, even with the learning rate α fixed.



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Linear Regression Model

$$h_{\mathbf{w}}(x) = w_0 + w_1 x$$

$$J(w_0, w_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right)^2$$



$$\begin{aligned}\frac{\partial}{\partial w_j} J(w_0, w_1) &= \frac{\partial}{\partial w_j} \frac{1}{2m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right)^2 \\ &= \frac{\partial}{\partial w_j} \frac{1}{2m} \sum_{i=1}^m \left(w_0 + w_1 x^{(i)} - y^{(i)} \right)^2\end{aligned}$$



repeat until convergence {

$$w_0 := w_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right)$$

$$w_1 := w_1 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)}$$

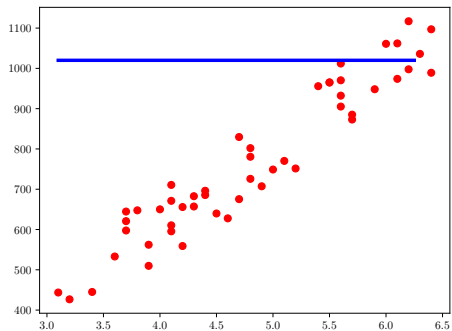
}

Gradient descent demonstration



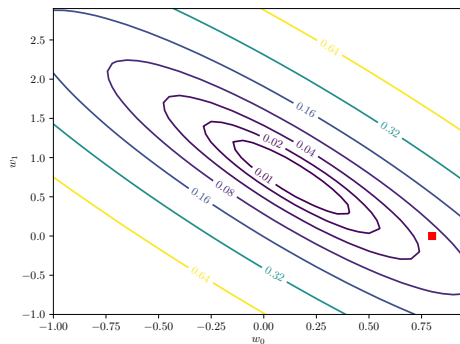
$$h_w(x)$$

For fixed w_0, w_1 , this is a function of x .



$$J(w_0, w_1)$$

Functions of the parameters w_0, w_1 .

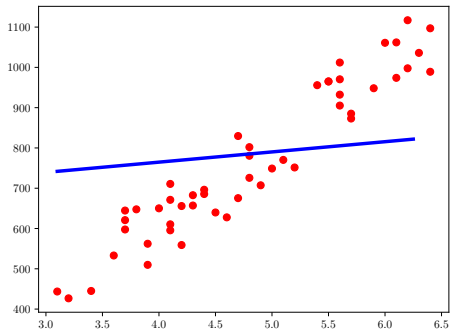


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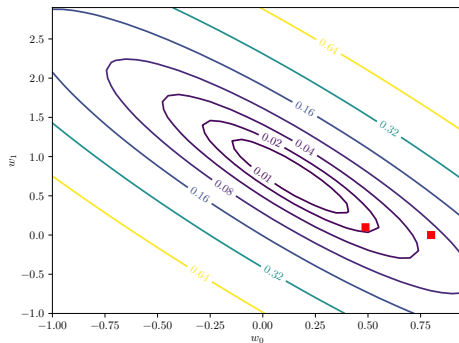
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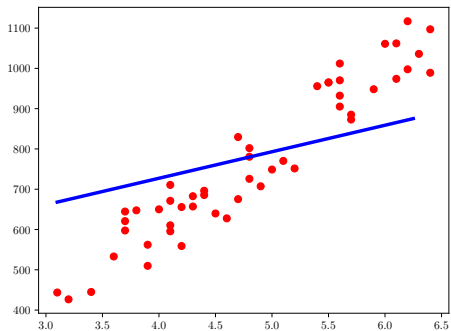


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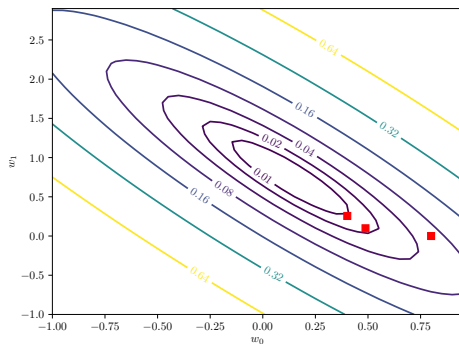
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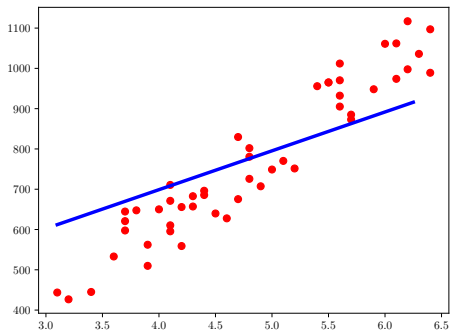


Gradient descent demonstration



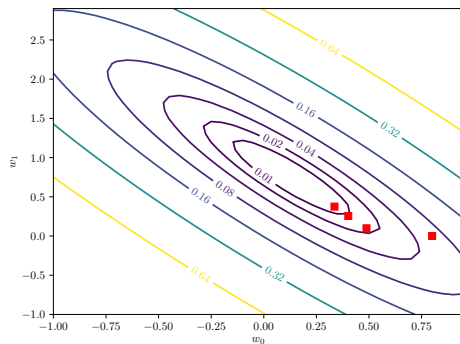
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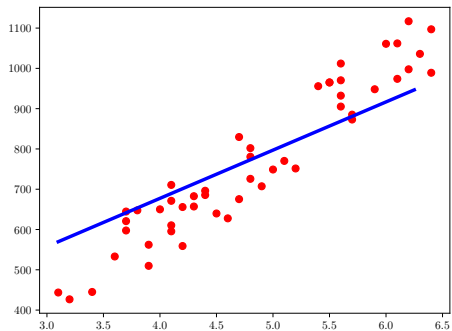


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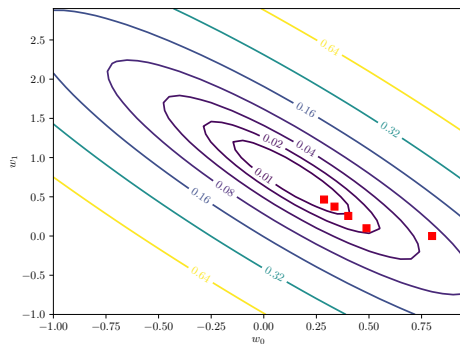
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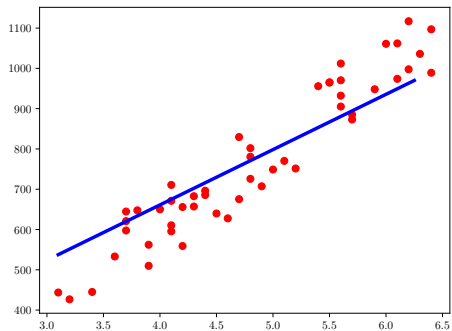


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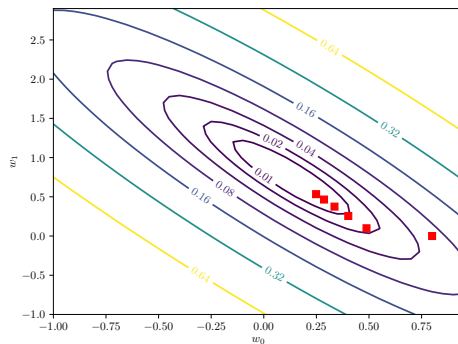
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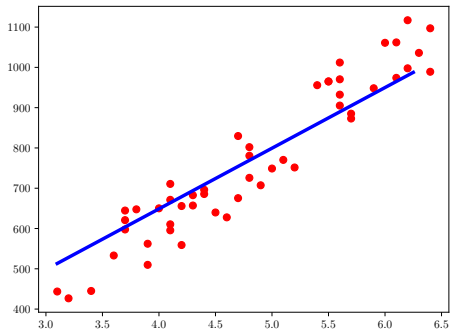


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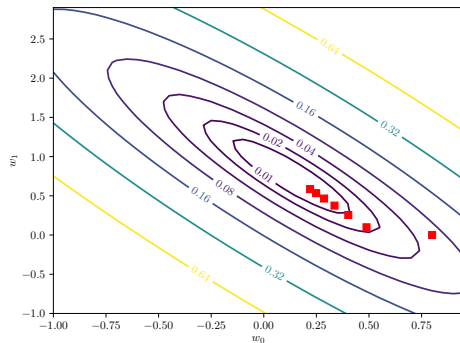
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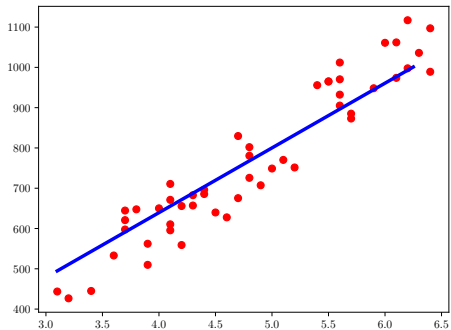


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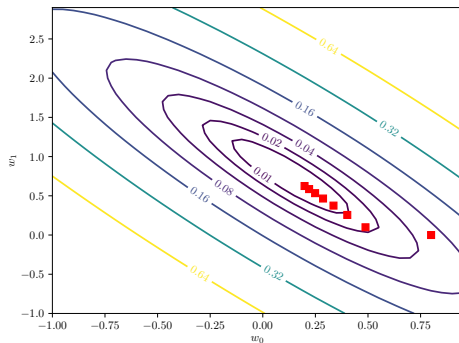
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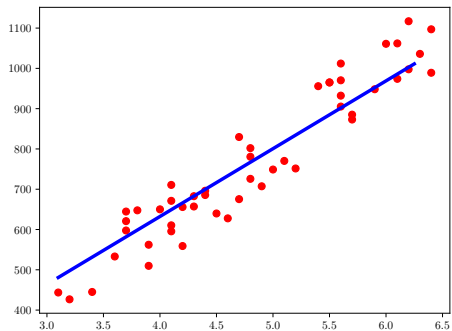


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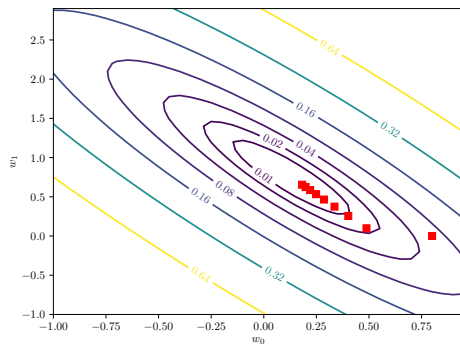
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Multivariate Linear Regression



Training set of phone prices

Display size	Price (USD)
x	y
3.6	499.7
4.1	570.5
4.2	551.9
5.5	918.9
6.2	980
$\vdots$	$\vdots$



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x	y
3.6	499.7
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- Hypothesis

$$h_{\mathbf{w}}(x) = w_0 + w_1x$$

- Cost function

$$J(w_0, w_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right)^2$$

- Goal

$$\arg \min_{w_0, w_1} J(w_0, w_1)$$

- Gradient descent

repeat until convergence {

$$w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(w_0, w_1), \quad j = 0, 1$$

}

Multiple Features (Variable)



Display size	CPU	MEM	Age	Price (USD)
x_1	x_2	x_3	x_4	y
4.1	1.1	24	3.5	570.5
5.5	1.2	2	1	918.9
6.2	1.2	26	2	980
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Notations:

- n number of features.
- $x^{(i)}$ input (features) of the i th training example.
- $x_j^{(i)}$ value of the j th feature in the i th training example.



- Previously

$$h_{\mathbf{w}}(x) = w_0 + w_1 x$$



- Previously

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x$$

- For the new phone price dataset

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$



- Previously

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x$$

- For the new phone price dataset

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

- Example:

$$h_{\mathbf{w}}(\mathbf{x}) = 100 + 0.2x_1 + 1.2x_2 + 0.01x_3 - 0.5x_4$$



$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$



$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

- For the convenience of notation, define $x_0 = 1$ ($x_0^{(i)} = 1$).

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n+1} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} \in \mathbb{R}^{n+1}$$



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$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n = \mathbf{w}^T \mathbf{x}$$

- Multivariate linear regression



- Hypothesis (Parameters $\mathbf{w} = [w_0, w_1, w_2, \dots, w_n]^T$).

$$h_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^T \mathbf{x} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$



- Hypothesis (Parameters $\mathbf{w} = [w_0, w_1, w_2, \dots, w_n]^T$).

$$h_{\mathbf{w}}(\mathbf{x}) = \mathbf{w}^T \mathbf{x} = w_0 + w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

- Cost function

$$J(w_0, w_1, w_2, \dots, w_n) = \frac{1}{2m} \sum_{i=1}^m (h_{\mathbf{w}}(\mathbf{x}^{(i)}) - y^{(i)})^2 \triangleq J(\vec{\mathbf{w}})$$



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- Gradient descent

repeat until convergence {

$$w_j := w_j - \alpha \frac{\partial}{\partial w_j} J(\mathbf{w})$$

(simultaneously update for every $j = 0, 1, 2, \dots, n$)

}



Previously ($n = 1$)

Repeat {

$$w_0 := w_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right)$$

$$w_1 := w_1 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)}$$

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}

New algorithm ($n \geq 1$)

Repeat {

$$w_j := w_j - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_w(x^{(i)}) - y^{(i)} \right) \cdot x_j^{(i)}$$

}



Previously ($n = 1$)

Repeat {

$$w_0 := w_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right)$$

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}

$$\frac{\partial}{\partial w_j} J(\mathbf{w})$$

Normal Equation



	Display size	CPU	MEM	Age	Price (USD)
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$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

$$\mathbf{w} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

Features and Generalized Linear Regression

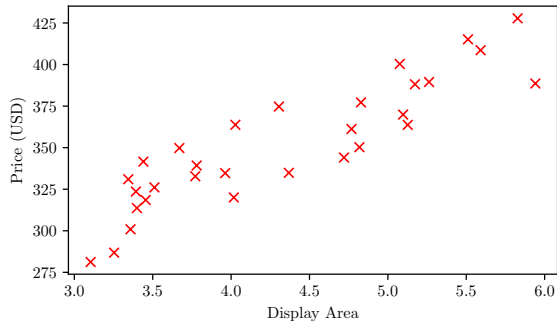


$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 \text{width} + w_2 \text{height}$$

$$x = \text{DisplayArea} = \text{width} \times \text{height}$$

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x$$

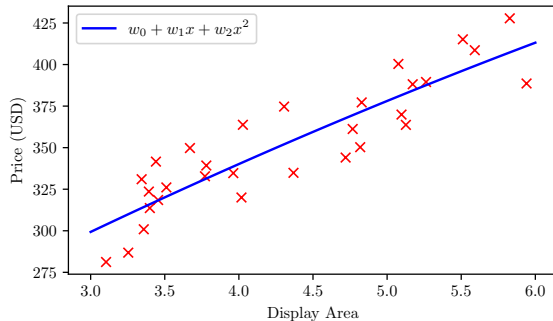
Polynomial Regression



$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1$$
$$= w_0 + w_1 (\text{size})$$

- $x_1 = \text{size}$

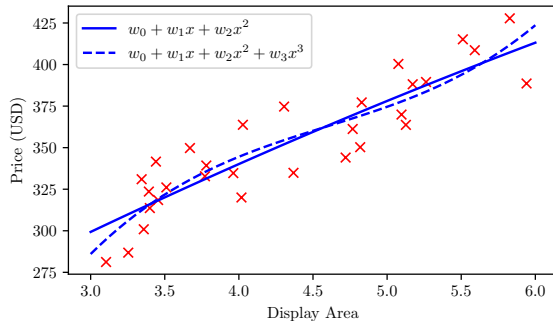
Polynomial Regression



$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2$$
$$= w_0 + w_1(\text{size}) + w_2(\text{size})^2$$

- $x_1 = \text{size}$
- $x_2 = (\text{size})^2$

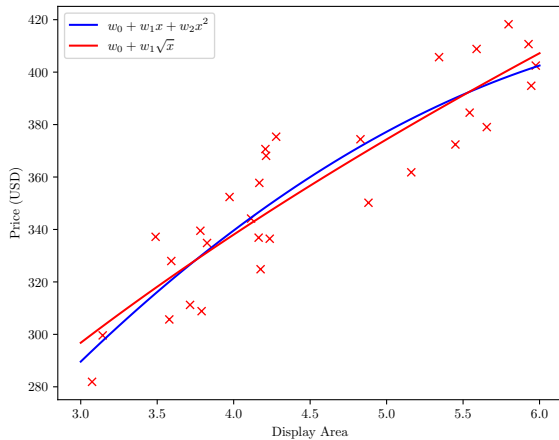
Polynomial Regression



$$\begin{aligned}h_{\mathbf{w}}(\mathbf{x}) &= w_0 + w_1x_1 + w_2x_2 + w_3x_3 \\ &= w_0 + w_1(\text{size}) \\ &\quad + w_2(\text{size})^2 + w_3(\text{size})^3\end{aligned}$$

- $x_1 = \text{size}$
- $x_2 = (\text{size})^2$
- $x_3 = (\text{size})^3$

Feature Choice



$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1(\text{size}) + w_2(\text{size})^2$$

$$h_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 \sqrt{\text{size}}$$

Polynomial series

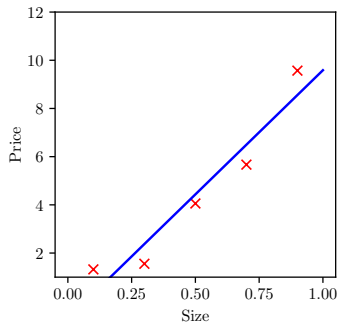
$$\mathbf{F} = \begin{bmatrix} x \\ x^2 \\ x^3 \\ \vdots \\ x^n \end{bmatrix}$$

General functions

$$\Phi(\mathbf{x}) = \begin{bmatrix} \varphi_1(\mathbf{x}) \\ \varphi_2(\mathbf{x}) \\ \vdots \\ \varphi_n(\mathbf{x}) \end{bmatrix}$$

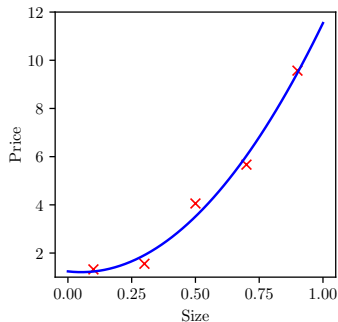
Regularization

Bias and Overfitting



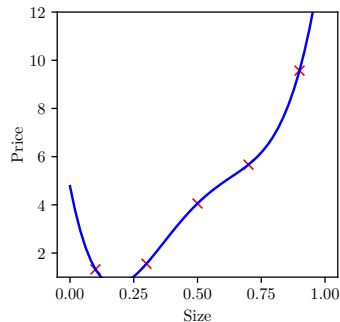
$$w_0 + w_1x$$

Underfit High bias



$$w_0 + w_1x + w_2x^2$$

Just right



$$\sum_{i=0}^4 w_i x^i$$

Overfit High Variance



$$J(\mathbf{w}) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\mathbf{w}}(\mathbf{x}^{(i)}) - y^{(i)})^2 + \right]$$



$$J(\mathbf{w}) = \frac{1}{2m} \left[\sum_{i=1}^m (h_{\mathbf{w}}(\mathbf{x}^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n w_j^2 \right]$$



Repeat {

$$w_0 := w_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right) x_0^{(i)}$$

$$w_j := w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right) x_j^{(i)} + \frac{\lambda}{m} w_j \right], \quad j = 1, 2, 3, \dots, n$$

}



Repeat {

$$w_0 := w_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right) x_0^{(i)}$$

$$w_j := w_j - \alpha \left[\frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right) x_j^{(i)} + \frac{\lambda}{m} w_j \right], \quad j = 1, 2, 3, \dots, n$$

}

$$w_j := w_j \left(1 - \alpha \frac{\lambda}{m} \right) - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\mathbf{w}}(x^{(i)}) - y^{(i)} \right) x_j^{(i)}$$

$$\mathbf{w} = \left(\mathbf{X}^T \mathbf{X} + \lambda \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{bmatrix} \right)^{-1} \mathbf{X}^T \mathbf{y}$$

Bayesian Linear Regression



- The error function of linear regression could be motivated as the maximum likelihood solution under an assumed Gaussian noise model.
- Assume that the target variable t is given by a deterministic function $y(\mathbf{x}, \mathbf{w})$ with additive Gaussian noise so that

$$t = y(\mathbf{x}, \mathbf{w}) + \varepsilon$$

where ε is a zero mean Gaussian random variable with precision (inverse of variance) β .

- Thus we can write

$$p(t|\mathbf{x}, \mathbf{w}, \beta) = \mathcal{N}(t|y(\mathbf{x}, \mathbf{w}), \beta^{-1})$$



- Now consider a data set of inputs $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ with corresponding target values $\{t_1, \dots, t_N\}$.
- Let us group the target variables $\{t_n\}$ into a column vector $\mathbf{t}$.
- We obtain the following expression for the likelihood function

$$p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \prod_{n=1}^N \mathcal{N}(t_n | y(\mathbf{x}_n, \mathbf{w}), \beta^{-1})$$



- Taking the logarithm of the likelihood function,

$$\begin{aligned}\ln p(\mathbf{t}|\mathbf{w}, \beta) &= \sum_{n=1}^N \ln \mathcal{N}(t_n|y(\mathbf{x}_n, \mathbf{w}), \beta^{-1}) \\ &= \frac{N}{2} \ln \beta - \frac{N}{2} \ln(2\pi) - \beta E_D(\mathbf{w})\end{aligned}$$

- where the **sum-of-squares error function** is defined by

$$E_D(\mathbf{w}) = \frac{1}{2} \sum_{n=1}^N [t_n - y(\mathbf{x}_n, \mathbf{w})]^2 = \frac{1}{2} \sum_{n=1}^N [t_n - \mathbf{w}^T \boldsymbol{\varphi}(\mathbf{x}_n)]^2.$$

The Bias-Variance Decomposition



- With squared loss function, the optimal prediction is given by the conditional expectation $h(\mathbf{x})$,

$$h(\mathbf{x}) = \mathbb{E}[t|\mathbf{x}] = \int t \cdot p(t|\mathbf{x}) dt.$$

- The expected squared loss can be written in the form

$$\mathbb{E}[L] = \int \left[y(\mathbf{x}) - h(\mathbf{x}) \right]^2 p(\mathbf{x}) d\mathbf{x} + \int \left[h(\mathbf{x}) - t \right]^2 p(\mathbf{x}, t) d\mathbf{x} dt.$$

- The second term, which is independent of $y(\mathbf{x})$, arises from the intrinsic noise on the data and represents the minimum achievable value of the expected loss.

The Bias-Variance Decomposition



- The first term depends on our choice for the function $y(\mathbf{x})$.
 - In practice, the data set $\mathcal{D}$ containing only a finite number N of data points.
- Consider the integrand of the first term, which for a particular data set $\mathcal{D}$ takes the form $[y(\mathbf{x}; \mathcal{D}) - h(\mathbf{x})]^2$.
- If we add and subtract the quantity $\mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})]$, and then expand:

$$\begin{aligned} & \left[y(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] + \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] - h(\mathbf{x}) \right]^2 \\ &= \left[y(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] \right]^2 + \left[\mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] - h(\mathbf{x}) \right]^2 \\ & \quad + 2 \left[y(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] \right] \left[\mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] - h(\mathbf{x}) \right]. \end{aligned}$$



- We now take the expectation of this expression w.r.t. $\mathcal{D}$ and note that the final term will vanish, giving

$$\begin{aligned} & \mathbb{E}_{\mathcal{D}} \left[y(\mathbf{x}; \mathcal{D}) - h(\mathbf{x}) \right]^2 \\ &= \underbrace{\left[\mathbb{E}_{\mathcal{D}} [y(\mathbf{x}; \mathcal{D})] - h(\mathbf{x}) \right]^2}_{(\text{bias})^2} + \underbrace{\mathbb{E}_{\mathcal{D}} \left[y(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}} [y(\mathbf{x}; \mathcal{D})] \right]^2}_{\text{variance}}. \end{aligned}$$

The Bias-Variance Decomposition



- If we substitute this expansion back into $\mathbb{E}(L)$, we obtain the following decomposition of the expected squared loss

$$\text{expected loss} = (\text{bias})^2 + \text{variance} + \text{noise}$$

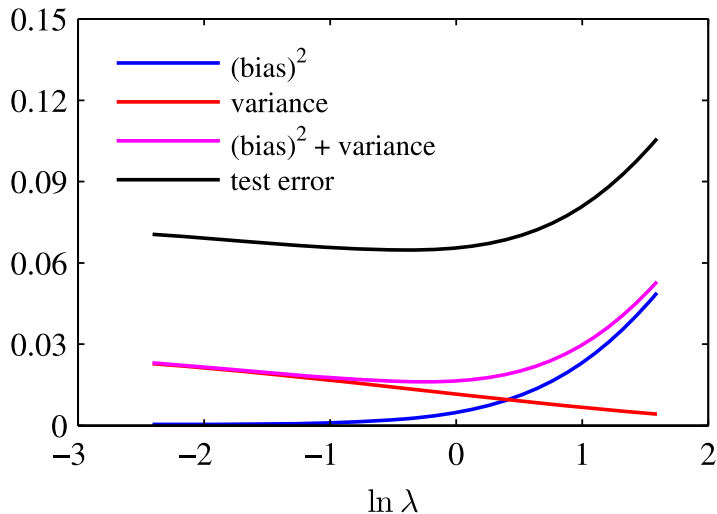
where

$$(\text{bias})^2 = \int \left[\mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] - h(\mathbf{x}) \right]^2 p(\mathbf{x}) d\mathbf{x}$$

$$\text{variance} = \int \mathbb{E}_{\mathcal{D}} \left[\left[y(\mathbf{x}; \mathcal{D}) - \mathbb{E}_{\mathcal{D}}[y(\mathbf{x}; \mathcal{D})] \right]^2 \right] p(\mathbf{x}) d\mathbf{x}$$

$$\text{noise} = \int [h(\mathbf{x}) - t]^2 p(\mathbf{x}, t) d\mathbf{x} dt$$

The Bias-Variance Decomposition



Our goal is to minimize the expected loss, which we have decomposed into the sum of a (squared) bias, a variance, and a constant noise term.



- We have discussed maximum likelihood for setting the parameters of a linear regression model.
- The effective model complexity, governed by the number of basis functions.
- The regularization coefficient controls the effective model complexity.
- How to decide the appropriate model complexity for the particular problem?
 - Why it cannot be decided by maximizing the likelihood function?



- We begin Bayesian treatment of linear regression by introducing a prior probability distribution over the model parameters $\mathbf{w}$.
- Note that $p(\mathbf{t}|\mathbf{w})$ is the exponential of a quadratic function of $\mathbf{w}$.
- The corresponding conjugate prior is therefore given by a Gaussian distribution

$$p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_0, \mathbf{S}_0)$$



- Let us compute the posterior distribution, which is proportional to the product of the likelihood function and the prior.
- Due to the choice of a conjugate Gaussian prior distribution, the posterior will also be Gaussian.
- The posterior distribution is in the form

$$p(\mathbf{w}|\mathbf{t}) = \mathcal{N}(\mathbf{w}|\mathbf{m}_N, \mathbf{S}_N)$$

where

$$\mathbf{m}_N = \mathbf{S}_N(\mathbf{S}_0^{-1}\mathbf{m}_0 + \beta\Phi^T\mathbf{t}), \quad \mathbf{S}_N^{-1} = \mathbf{S}_0^{-1} + \beta\Phi^T\Phi.$$



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- The maximum posterior weight vector is simply given by $\mathbf{w}_{\text{MAP}} = \mathbf{m}_N$.



- For simplicity, let us consider a particular form of Gaussian prior

$$p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I})$$

- The corresponding posterior distribution over $\mathbf{w}$ is then

$$\mathbf{m}_N = \beta \mathbf{S}_N \boldsymbol{\Phi}^T \mathbf{t}$$

$$\mathbf{S}_N^{-1} = \alpha \mathbf{I} + \beta \boldsymbol{\Phi}^T \boldsymbol{\Phi}.$$

- Consider an infinitely broad prior $\mathbf{S}_0 = \alpha^{-1}\mathbf{I}$ with $\alpha \rightarrow 0$, the mean $\mathbf{m}_N$ reduces to the maximum likelihood value $\mathbf{w}_{\text{ML}}$.

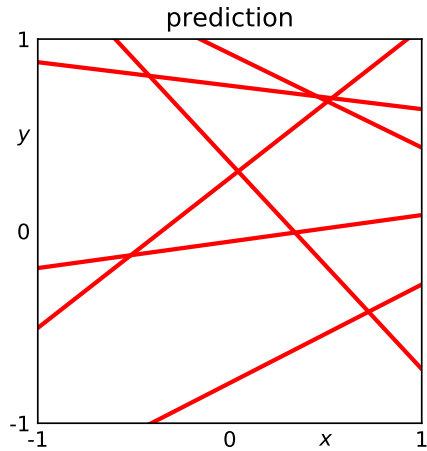
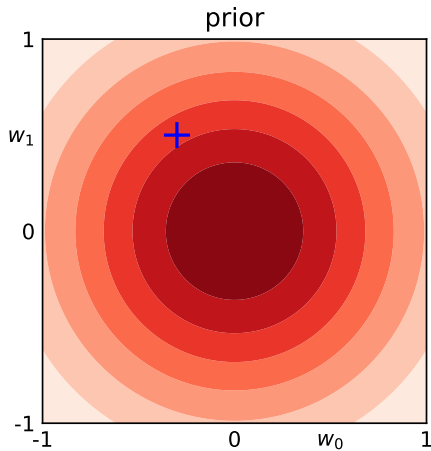


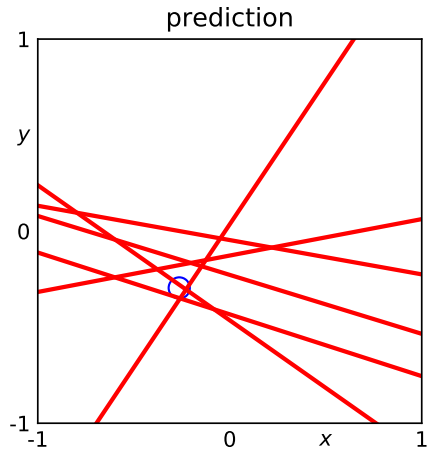
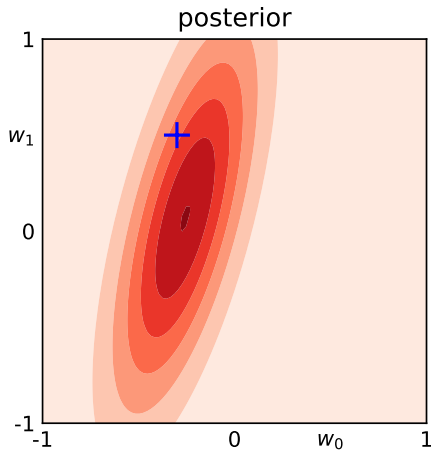
- The log of the posterior distribution is given by the sum of the log likelihood and the log of the prior

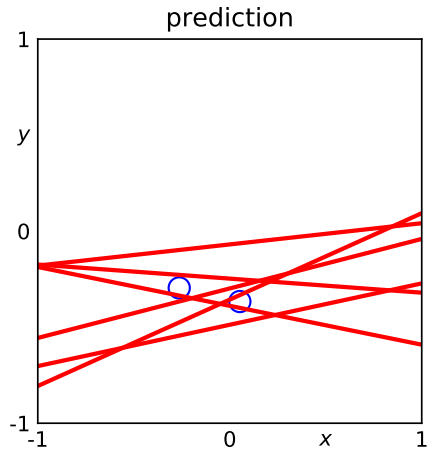
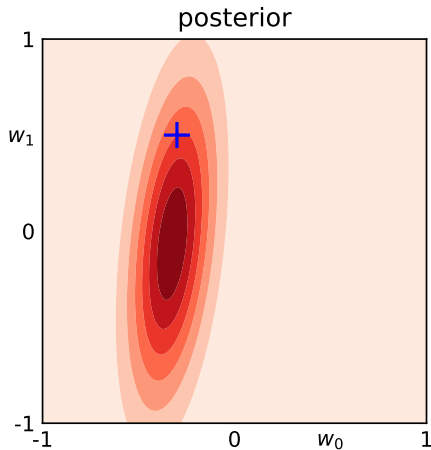
$$\ln p(\mathbf{w}|\mathbf{t}) = -\frac{\beta}{2} \sum_{n=1}^N [t_n - \mathbf{w}^T \boldsymbol{\varphi}(\mathbf{x}_n)]^2 - \frac{\alpha}{2} \mathbf{w}^T \mathbf{w} + \text{const.}$$

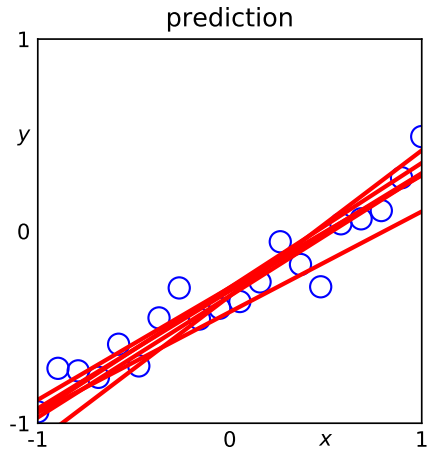
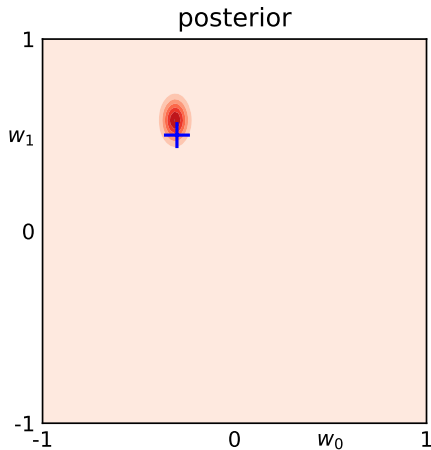
- Maximization of this posterior distribution is equivalent to the minimization of the regularized sum-of-squares error function with $\lambda = \alpha/\beta$.

Parameter distribution











- In practice, we are not usually interested in the value of $\mathbf{w}$ itself
- but rather in making predictions of t for new values of $\mathbf{x}$.
- This requires that we evaluate the **predictive distribution** defined by

$$p(t|\mathbf{t}, \alpha, \beta) = \int p(t|\mathbf{w}, \beta)p(\mathbf{w}|\mathbf{t}, \alpha, \beta)d\mathbf{w}$$

which involves the convolution of two Gaussian distributions.



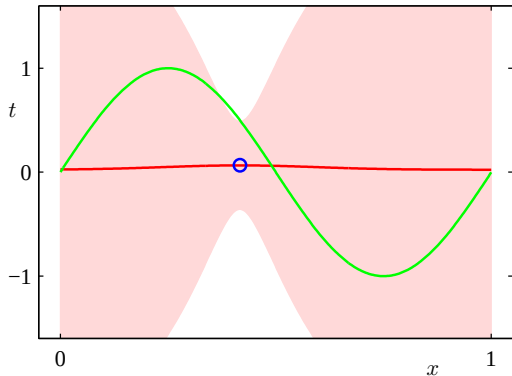
- The predictive distribution takes the form

$$p(t|\mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}\left(t | \mathbf{m}_N^\top \boldsymbol{\Phi}(\mathbf{x}), \sigma_N^2(\mathbf{x})\right)$$

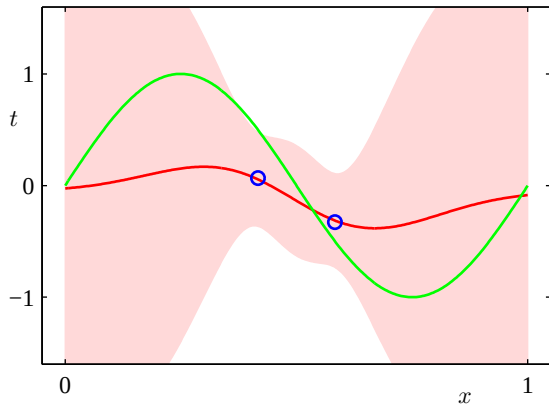
- where the variance of the predictive distribution is given by

$$\sigma_N^2 = \frac{1}{\beta} + \boldsymbol{\Phi}(\mathbf{x})^\top \mathbf{S}_N \boldsymbol{\Phi}(\mathbf{x})$$

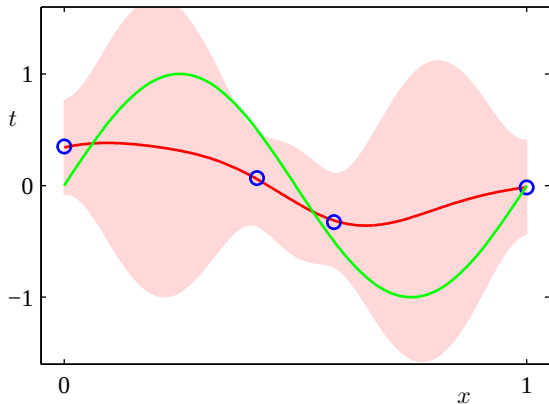
- The first term represents the noise on the data
- the second term reflects the uncertainty associated with the parameters $\mathbf{w}$.
- Because the noise process and the distribution of $\mathbf{w}$ are independent Gaussians, their variances are additive.



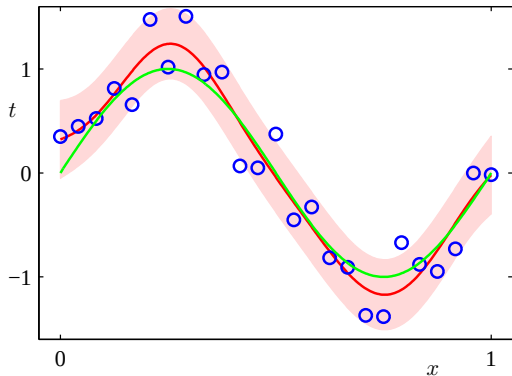
- Examples of the predictive distribution for a model consisting of 9 Gaussian basis functions
- Data sets of size $N = 1, 2, 4$, and 25 are shown in the plots by the blue circles.



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- As additional data points are observed, the posterior distribution becomes narrower.



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- Data sets of size $N = 1, 2, 4$, and 25 are shown in the plots by the blue circles.
- In the limit $N \rightarrow \infty$, the second term goes to zero, and the variance of the predictive distribution arises solely from the additive noise governed by the parameter β .